

PROGRAMMING BASICS

DAY 1

1. Arithmetic Operations

Problem:

Given two integers a and b, print their sum, difference, product, and quotient.

Input:

Two integers a b

Output:

Print 4 values: sum, difference, product, quotient

2. Rectangle Properties

Problem:

Given length l and breadth b of a rectangle, find its **area** and **perimeter**.

Input:

Two integers l b

Output:

Area and perimeter

3. Circle Calculations

Problem:

Given radius r, find diameter, circumference, and area of the circle.

Input:

One integer r

Output:

Diameter, circumference, area
(Use $\pi = 3.14$)

4. Length Conversion

Problem:

Convert a given length in centimeters into meters and kilometers.

Input:

One integer cm

Output:

Meters and kilometers

5. Days Conversion**Problem:**

Convert given number of days into years, weeks, and remaining days.

Input:

One integer days

Output:

Years, weeks, remaining days

6. Triangle Area**Problem:**

Given base b and height h, find area of triangle.

Input:

Two integers b h

Output:

Area

7. Equilateral Triangle Area**Problem:**

Given side a, find area of equilateral triangle.

Formula:

$$\text{Area} = (\sqrt{3} / 4) \times a^2$$

Input:

One integer a

Output:

Area

8. Marks Calculation**Problem:**

Given marks of 5 subjects, calculate total, average, and percentage.

Input:

Five integers

Output:

Total, average, percentage

9. Swap using Third Variable**Problem:**

Swap two numbers using a third variable.

Input:

Two integers a b

Output:

Swapped values

10. Swap without Third Variable**Problem:**

Swap two numbers without using a third variable.

Input:

Two integers a b

Output:

Swapped values

11. Sum of Two Numbers**Problem:**

Given two integers a and b, print their sum.

Input:

Two integers a b

Output:

Single integer $\rightarrow$ sum of a and b

12. Area of Circle**Problem:**

Given radius r, calculate the area of a circle.

Input:

One floating-point number r

Output:

Area of the circle (use $\pi = 3.14$ or built-in value)

13. Simple Interest**Problem:**

Given principal p , rate r , and time t , calculate simple interest.

Input:

Three floating-point numbers p r t

Output:

Simple interest

14. Even or Odd**Problem:**

Given an integer n , determine whether it is even or odd.

Input:

One integer n

Output:

Print "Even" or "Odd"

15. Maximum of Two Numbers**Problem:**

Given two integers a and b , print the larger number.

Input:

Two integers a b

Output:

Maximum value

16. ASCII Value**Problem:**

Given a character c , print its ASCII value.

Input:

One character c

Output:

Integer ASCII value

17. Celsius to Fahrenheit

Problem:

Convert temperature from Celsius to Fahrenheit.

Input:

One floating-point number c

Output:

Temperature in Fahrenheit

18. Check Alphabet

Problem:

Given a character c, check whether it is an alphabet.

Input:

One character

Output:

"Alphabet" or "Not Alphabet"

19. Sum of Digits (2-digit)

Problem:

Given a 2-digit number, find the sum of its digits.

Input:

One integer n

Output:

Sum of digits

20. Reverse a 3-digit Number

Problem:

Given a 3-digit number, reverse it.

Input:

One integer n

Output:

Reversed number

21. Power Calculation

Problem:

Given two integers a and b, compute a^b .

Input:

Two integers a b

Output:

Result of a raised to power b

22. Simple Calculator

Problem:

Given two integers and an operator (+, -, *, /), perform the operation.

Input:

a b op

Output:

Result of operation

23. Number Sign

Problem:

Determine whether a number is positive, negative, or zero.

Input:

One integer n

Output:

Positive, Negative, or Zero

24. Perimeter of Rectangle

Problem:

Given length l and width w, find perimeter.

Input:

Two integers l w

Output:

Perimeter

25. Average of Three Numbers

Problem:

Calculate average of three numbers.

Input:

Three integers a b c

Output:

Average value

26. Divisibility Check**Problem:**

Check whether a number is divisible by both 5 and 11.

Input:

One integer n

Output:

Yes or No

27. Postfix & Prefix Output**Problem:**

Given a = 5, print results of a++ and ++a.

Input:

No input

Output:

Two values showing postfix and prefix behavior

28. Operator Precedence**Problem:**

Evaluate the expression:

$10 + 5 * 2$

Input:

No input

Output:

Result of expression

29. Evaluate Complex Expression**Problem:**

Given three integers a, b, and c, evaluate the expression:

$$\text{result} = a^2 + b^2 + 2ab + c/2$$

Input:

Three integers a b c

Output:

Value of the expression

30. Sum of Three Numbers without Using +**Problem:**

Given three integers a, b, and c, find their sum without using the + operator.

Input:

Three integers a b c

Output:

Sum of the three numbers

Hint:

Use bitwise operators (^, &, <<)